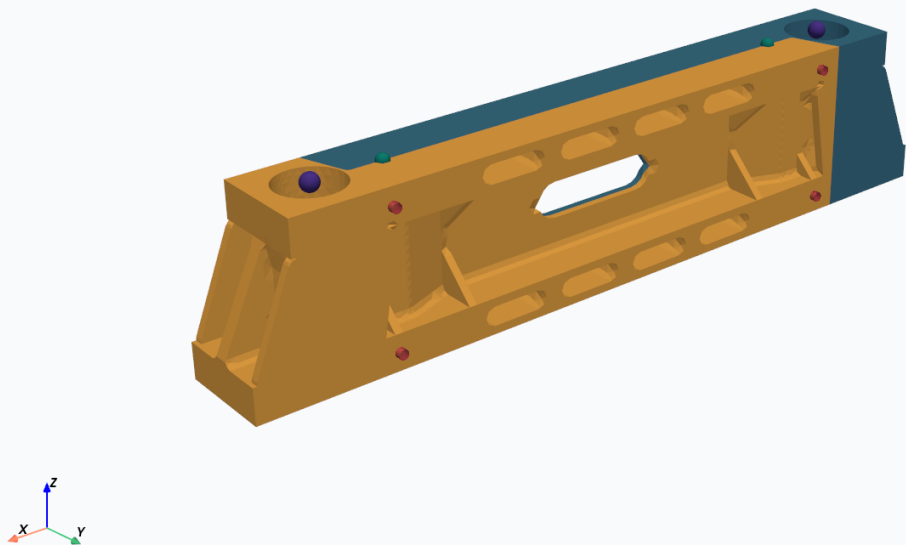


Selected-run evidence status: completed_unassessed

Geometry

350 mm offset clamshell original assembly axes

- ▲ Lower shell
- ▲ Upper shell



Red: four closure-bolt axes Violet: motor seats Teal: cable channels

Recorded assembly geometry. Original geometry/assembly overview from this selected case's matched report.

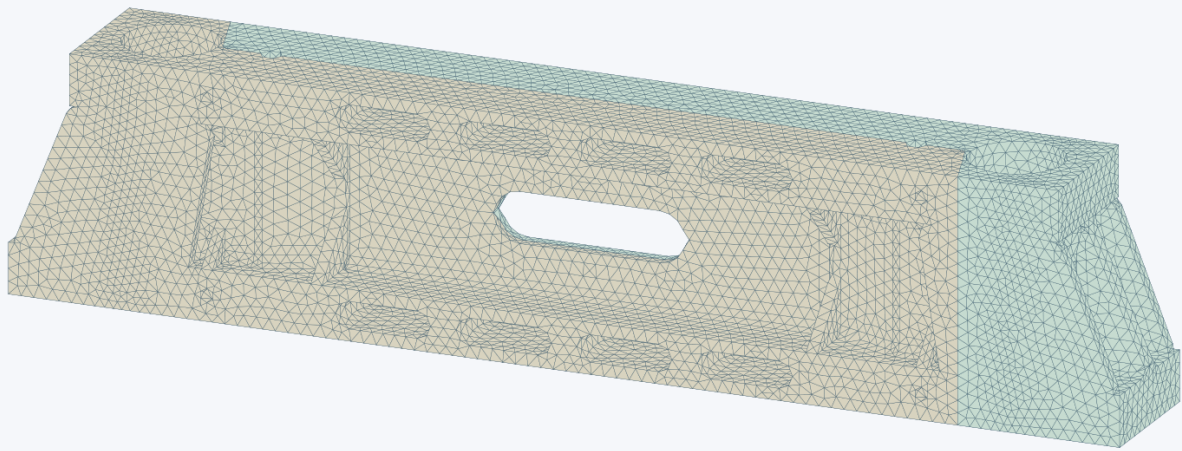
Mesh

Mesh overview and local detail

Mesh nodes	22673	
Tetrahedral elements	78906	
Meshed parts	2	
Nominal element size	4.0	mm
Minimum mean-ratio quality	0.0853	0-1

Mesh overview

Undeformed exterior mesh

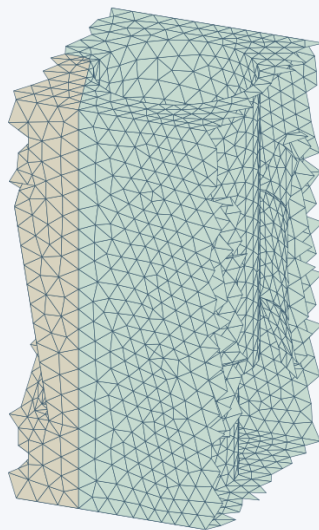


Exterior mesh faces • undeformed geometry

Mesh overview. Undeformed exterior topology of the selected case, verified against its recorded solved-mesh hashes.

Pad-root refinement detail

Pad-root region



Exterior mesh faces • undeformed geometry

Pad-root refinement detail. Positive pad-root region, $X = 112\text{--}150\text{ mm}$. Mesh metadata records a 0.65 target factor (2.6 mm local target).

Mean-ratio quality describes element shape and does not establish stress or time-step convergence.

Setup

Recorded selected-case model

Initial impact velocity	50	m/s
Requested simulation duration	8	ms
Foam depth	203.2	mm
Foam width	41.275	mm
Foam height	152.4	mm
Foam element size	10	mm
Initial contact gap	0.1	mm
Contact friction coefficient	0.1	dimensionless
Foam curve scale	1	dimensionless
Time-step scale	0.6	dimensionless
Requested state-output interval	0.01	ms
Startup mass-scaling DT2MS	-1.5e-07	s
Termination time-step ratio	0.01	dimensionless
Motor mass	0.1616	kg
Moving mass	0.7413749	kg
Pusher mass	0.5797749	kg
Initial kinetic energy	926.7187	J

Recorded material, contact and boundary assumptions

- Both offset clamshell shells are flexible; four local perfectly clamped closure-bolt collars. Seam contact allows opening elsewhere. No bolt preload, slip or failure qualification.
- Motor loaded upper-bound inertia distributed over actual bore nodes; no motor housing stiffness or exact inertia tensor.
- PA12-CF10 elastic isotropic proxy $E=1806.6\text{MPa}$ $\nu=.35$ density 1060; print +Z motor axes. No strength/failure claims.
- No fracture, erosion, motor thrust, carrier, released endcap or invented 35g hardware allowance.
- Rigid 8mm aluminum spreaders initially stationary; fixed x,y and rotations, free z without travel limit.

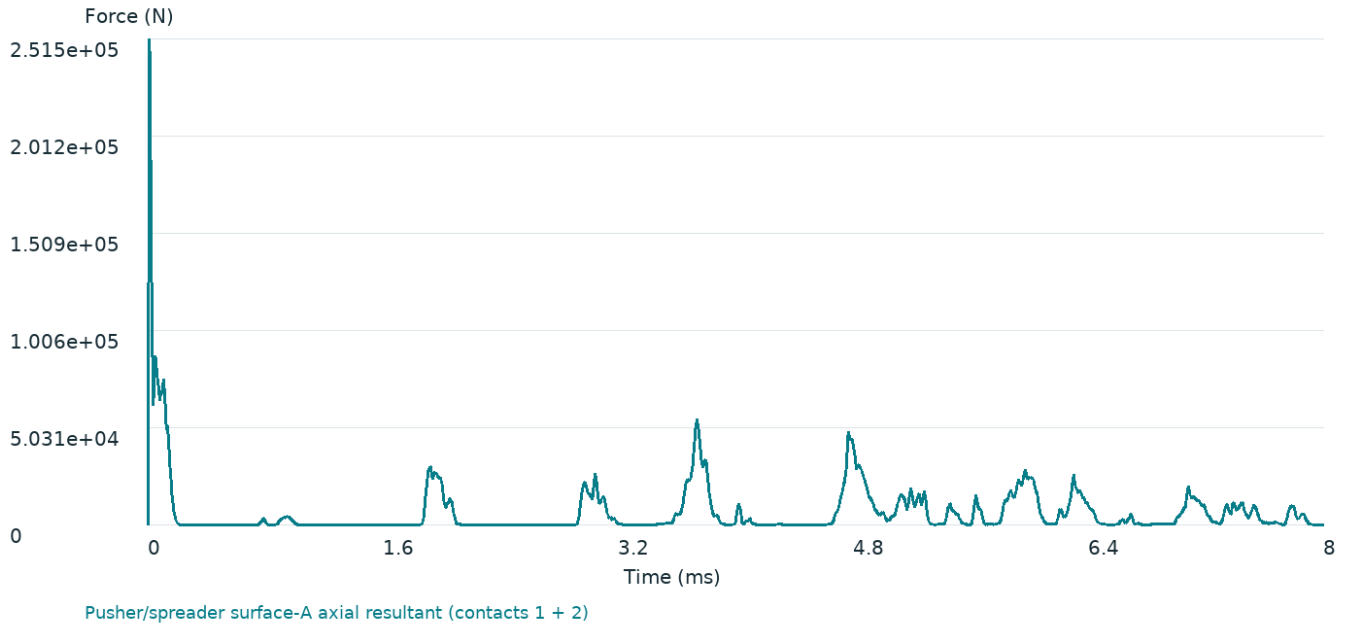
- Front and rear pusher bore-quadrant averages follow fixed zero-clearance coaxial guide axes through eight disjoint LINEAR_GLOBAL X/Y equations; every Z DOF is free. Cable compliance and clearance are idealized.
- All axial stopping reaction passes through transient contact; no axial guide constraint.
- Foam rear z fixed, x/y free: frictionless fixed backing; sides unconfined.
- MAT057 synthetic loading and unloading; HU=.05 BETA=0 DAMP=.1 SHAPE=1 FAIL=0. Not a product material model.
- Generic foam anchored at one catalog point; remaining curve chosen for nonbottoming comparative stop.
- Startup-only local mass scaling requested; added mass and inertia error MUST be assessed from solver output.
- Mesh, timestep, energy and momentum require independent assessment.

Solve

Recorded run status	completed_unassessed	
Recorded reached simulation time	7.99995	ms
Recorded requested simulation time	8	ms
Recorded solver wall time	3833.062	s
Recorded termination state	normal_termination_confirmed	

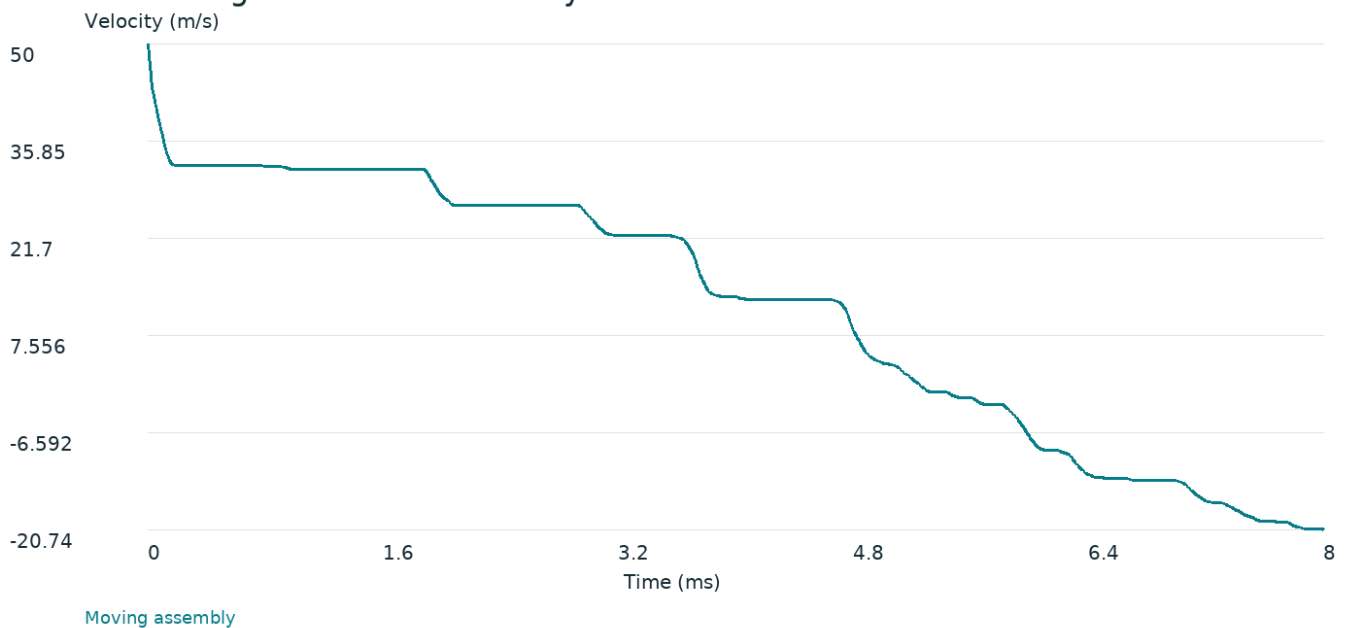
Separate solve charts

Contact force resultant



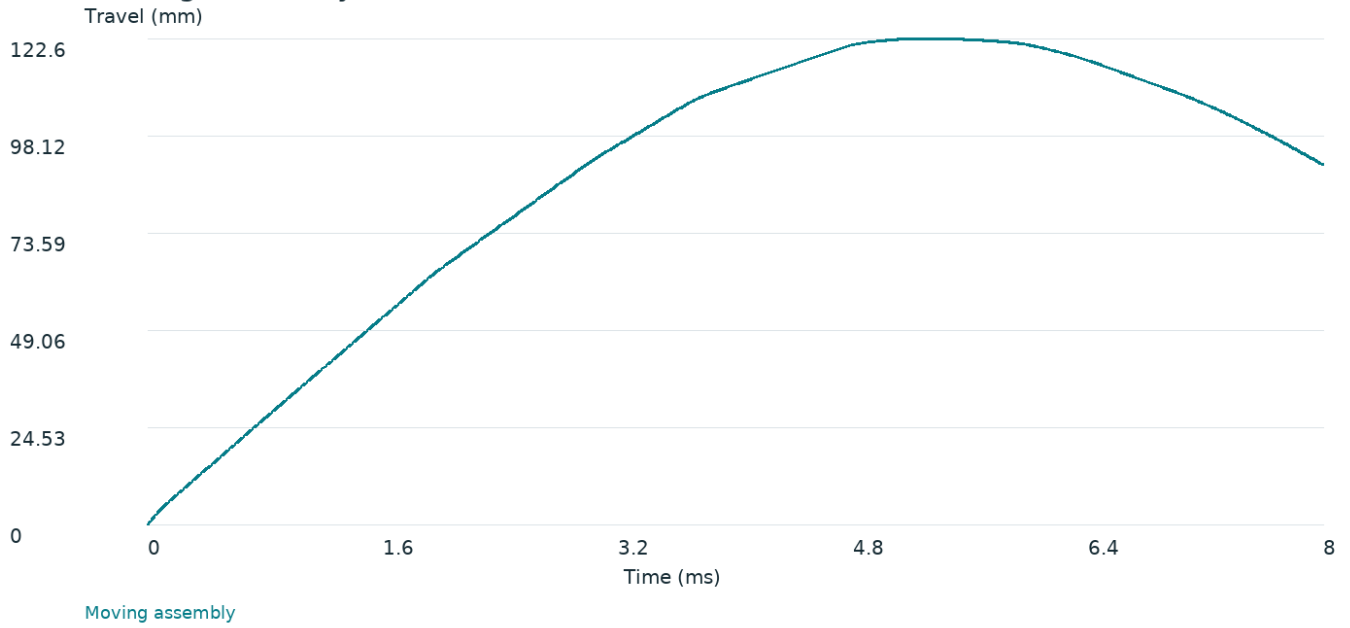
Contact force resultant. [projects/offset_clamshell_impact_20260922/cases/clamshell-4mm-v1/assessment/contact_histories.csv](#)

Mass-weighted forward velocity



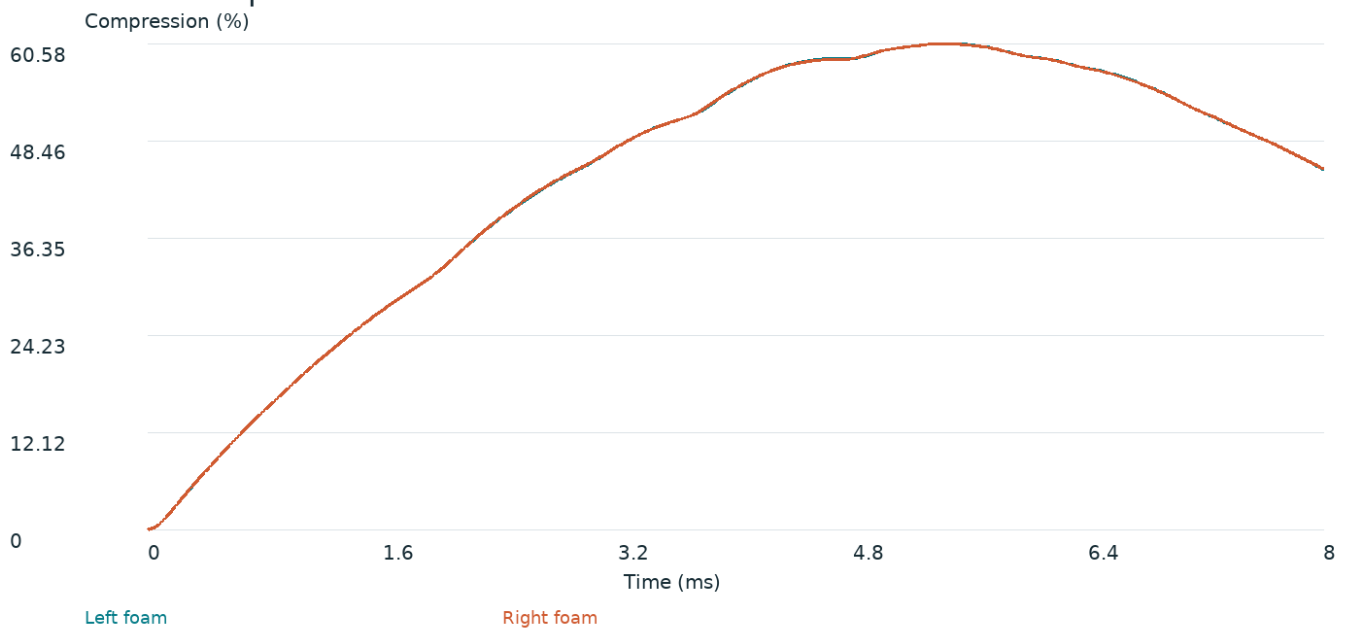
Mass-weighted forward velocity.
[projects/offset_clamshell_impact_20260922/cases/clamshell-4mm-v1/assessment/time_histories.csv](#)

Moving assembly travel



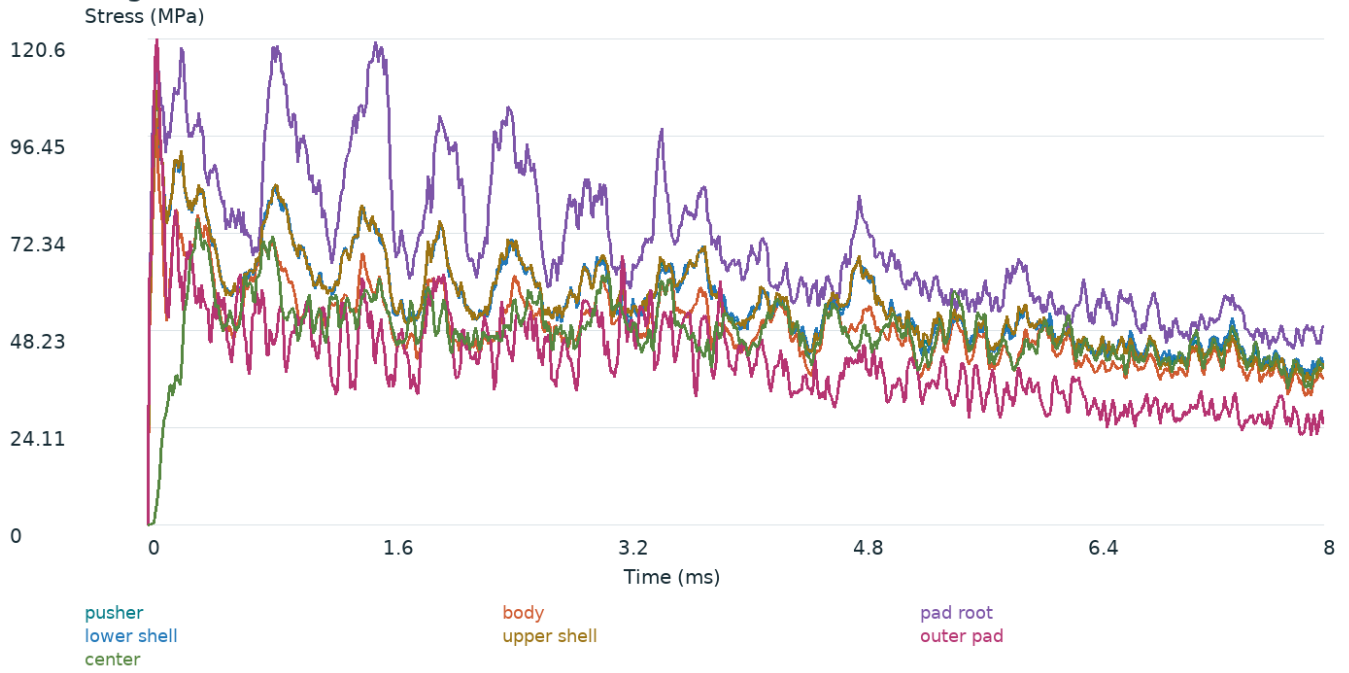
Moving assembly travel. projects/offset_clamshell_impact_20260922/cases/clamshell-4mm-v1/assessment/time_histories.csv

Foam compression



Foam compression. projects/offset_clamshell_impact_20260922/cases/clamshell-4mm-v1/assessment/time_histories.csv

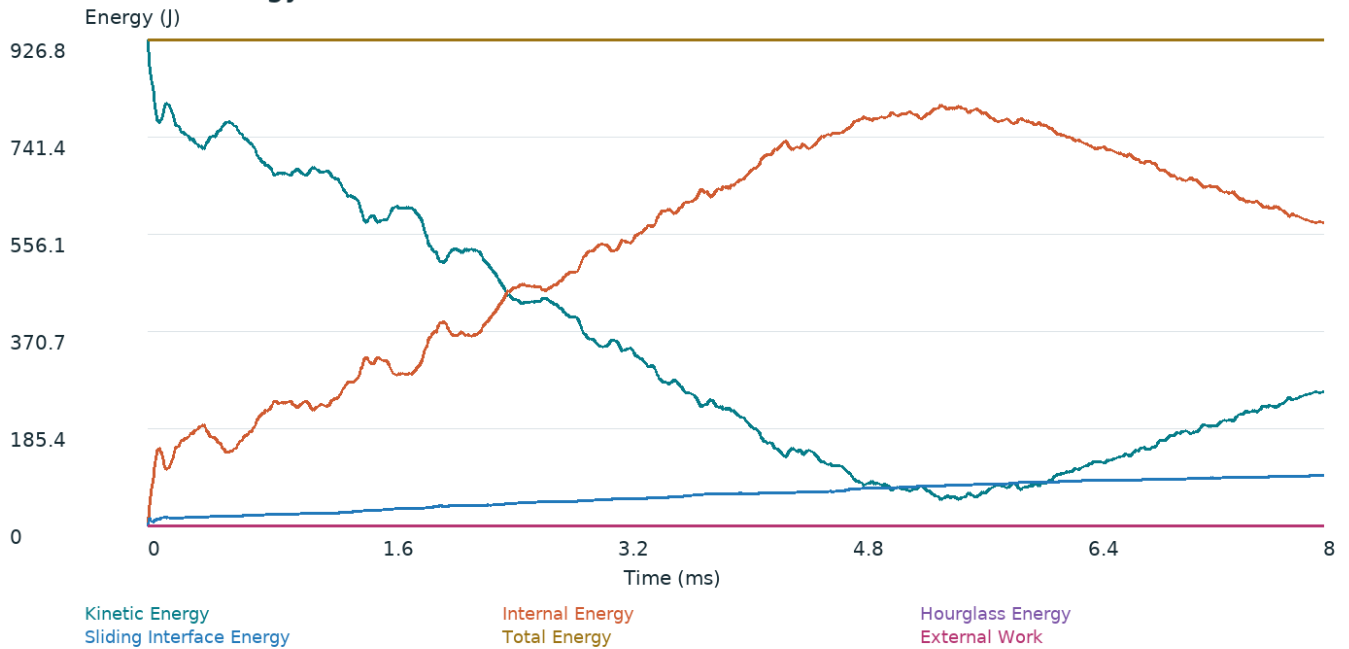
Regional P99 von Mises stress



Regional P99 von Mises stress.

projects/offset_clamshell_impact_20260922/cases/clamshell-4mm-v1/assessment/stress_histories.csv

Global energy



Global energy. projects/offset_clamshell_impact_20260922/cases/clamshell-4mm-v1/assessment/global_energy_histories.csv

Results

Offset-clamshell pusher: 50 m/s impact report

Candidate: supplied 350 mm Lower + Upper clamshell assembly. **Analysis:** ANSYS LS-DYNA explicit transient, two printed flexible shells, four ideal local closure joints, two motor masses, original two-spreader / eight-inch-foam fixture. **Observation:** 8.000 ms. **Status:** the completed 4 mm case passes the single-case numerical screens. The 4 → 3 mm candidate mesh screen **fails**. Failed measures: Bolt-collar mean relative motion.

The modeled pusher first stops at **5.273 ms after 122.65 mm**, with a **251.55 kN** short initial pusher-to-spreader force peak. Sampled central seam opening reaches **9.88 mm**; joint hardware capacity remains unqualified.

This is a comparative elastic demand study. It does not predict printed-part survival, bolt strength, preload, slip, physical rebound or commercial foam capacity.

1. Supplied geometry and assembly

The supplied assembly STEP has two valid solids. It was meshed as **two independent flexible parts** with their offset seam and inter-shell contact. The one-piece historical master was not used. Read-only native SolidWorks volumes and source hashes are in the CAD inspection and native verification. The native assembly and supplied STEP volumes differ by about 0.000053%; the native shell save times postdate the STEP exports, so feature-level revision identity cannot be proved from volume alone.

350 mm offset clamshell original assembly axes

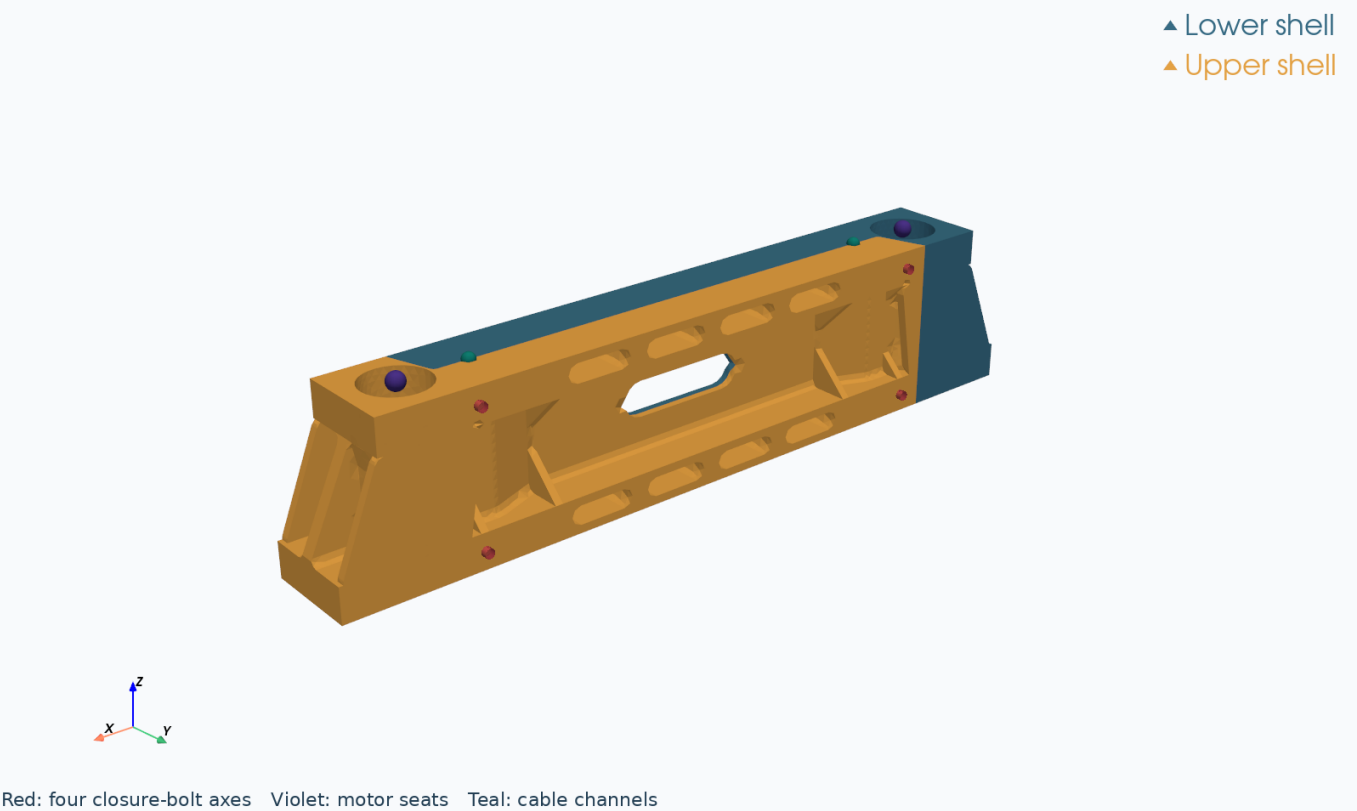


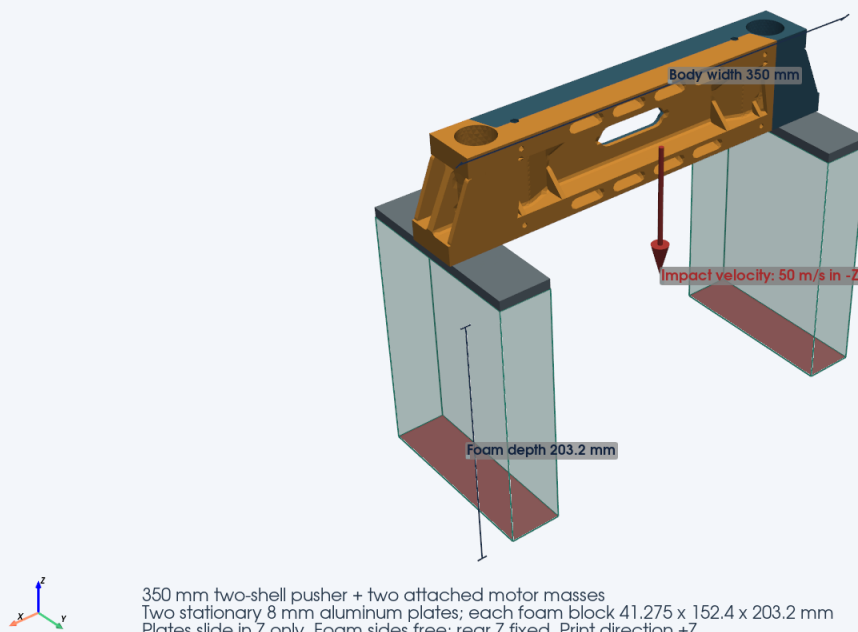
Figure 1. Original assembly axes. Red marks four closure-bolt axes, violet the two motor seats, and teal the cable channels. Lower and Upper are separate mesh parts.

Component / interface	Representation
Lower shell / Upper shell	First-order flexible tetrahedra; STEP volume 271.381 / 271.381 cm ³ ; FE mass 289.86 / 289.92 g
Closure joint	Four local ideal clamped collars at X = ±112.5 mm, Z = 7.9375 and 68.2625 mm; inter-shell frictional contact elsewhere
Motors	Two F67-9W masses at X = ±133.25 mm, 80.8 g each , attached to their respective shell bore nodes
Cable guidance	Front and rear transverse weighted-average bore constraints at X = ±101.6 mm; all axial Z degrees remain free
Excluded	Released carrier/end cap, historical removable cable cap, one-piece master, and unsupported 35 g hardware allowance

The printed CAD outer width is 350 mm. The unchanged spreaders reach to X = ±173.02 mm, about 1.98 mm short of each extreme. The actual meshed front-face overlap is **1,615.9 mm² left** and **1,615.9 mm² right**; the patch calculation clips exposed FE triangles against the fixed plate footprints. The setup audit verifies separate part node IDs, all joint memberships, motor locations, initial 0.1 mm impact clearance and engagement on both sides.

COMPLIANT STOP INITIAL MODEL GEOMETRY

- ▲ Foam
- ▲ Fixed rear faces
- ▲ Al spreaders
- ▲ Lower shell
- ▲ Upper shell



Initial assembly and fixed reference fixture

Figure 2. Undeformed candidate in the reference fixture. The aluminum spreaders translate axially; foam side faces are free and rear Z faces are fixed.

2. Physical model and dynamic load path

The pusher enters at **50 m/s in -Z**. Printed material uses density 1060 kg/m³, isotropic elastic $E = 1806.6$ MPa and assumed Poisson ratio 0.35. Printed shells weigh **579.77 g in the 4 mm FE mesh**; with motors, moving mass is **741.37 g** and nominal initial kinetic energy is **926.72 J**. The mesh's printed volume exceeds the STEP volume by about 0.8%; the exact per-shell errors remain in the setup audit.

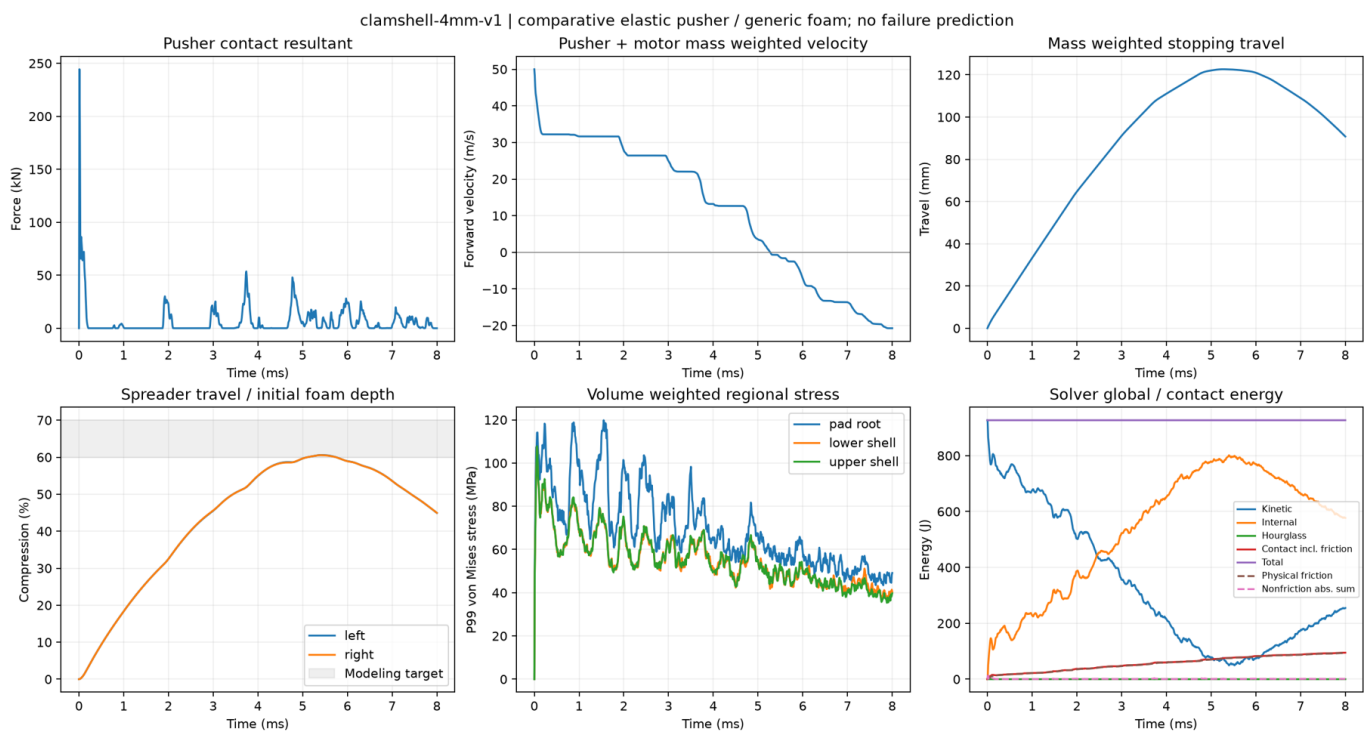
Each initially stationary aluminum spreader is 41.275 × 152.4 × 8 mm and 135.87 g. Two 203.2 mm unconfined foam blocks use the reference MAT057 nonlinear synthetic compression/unloading curve (54.46 kg/m³, 0.30337 MPa at 25% compression). Plate-to-foam friction is zero; pusher-to-plate and inter-shell friction are 0.1. The initial pad gap is 0.1 mm. No axial travel limit, ongoing motor thrust, motor casing stiffness, failure law or bolt preload is included. See the complete model and material curve and solved keyword deck.

Load path: moving shells and motors → two finite-mass spreaders → nonlinear foam → fixed rear backing. Force peaks at these locations occur at different times because the spreaders accelerate and the foam deforms.

3. Motion and stopping event

Event	Candidate result
First mass-weighted axial stop	5.273 ms
Travel at first stop	122.65 mm
Maximum moving travel	122.65 mm
Peak foam compression, left / right	60.58% / 60.55%
Final state at 8.000 ms	rebounding at 20.74 m/s
Largest nonrigid displacement / RMS	7.09 / 2.71 mm

“First stop” is the first zero crossing of mass-weighted axial pusher velocity. It does not mean every node is stationary. Compression is each spreader’s axial travel divided by initial foam depth; local foam strains differ.



Candidate time histories

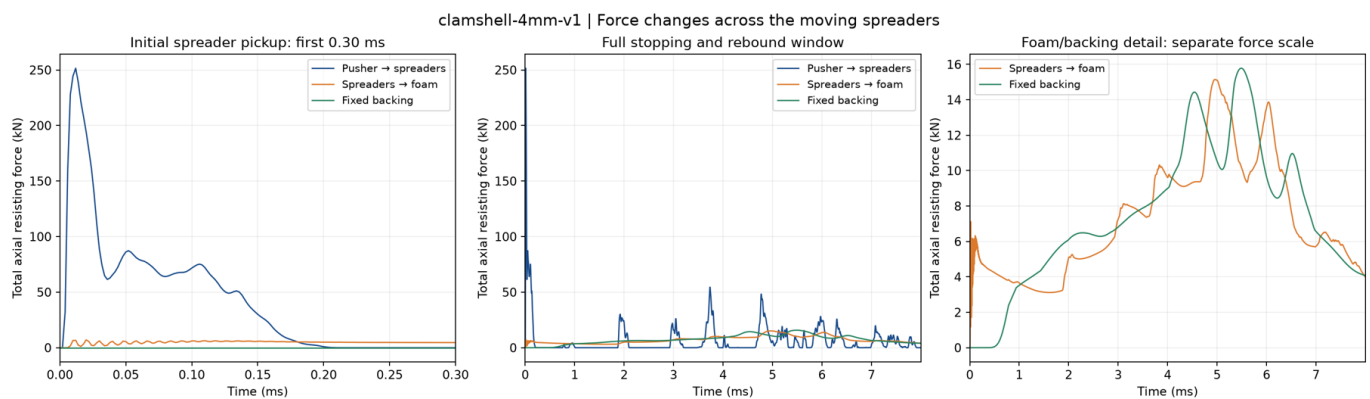
Figure 3. Saved-state speed, travel, compression, regional stress and energy. Contact load on this plot is sampled at binary times; Figure 4 retains native force sampling for the brief early peak.

Time	Evidence-based event
0.012 ms	Largest pusher-to-spreader force
0.380 ms	Central-region P95 stress reaches its own maximum
0.420 ms	Largest nonrigid shell displacement

Time	Evidence-based event
1.070 ms	Largest sampled central seam opening
5.273 ms	First mass-weighted axial stop
8.000 ms	End of requested observation window; rebounding at 20.74 m/s

4. Contact and backing forces

Location	Peak positive axial force	Peak time
Pusher → spreaders	251.55 kN	0.012 ms
Spreaders → foam	15.15 kN	4.960 ms
Fixed backing	15.78 kN	5.488 ms



Measured force path

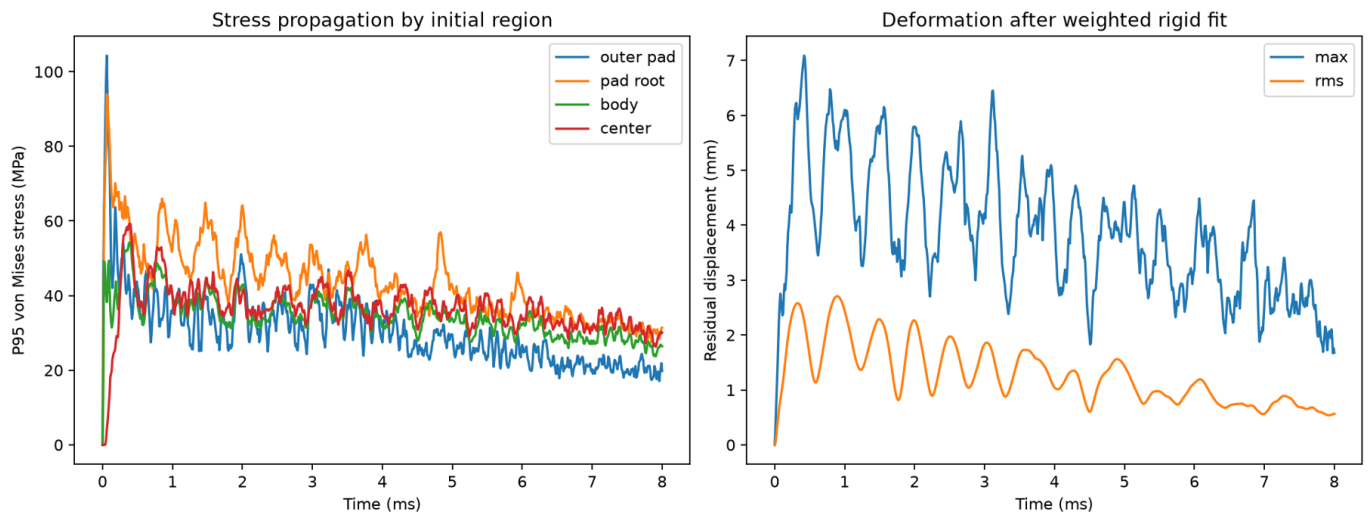
Figure 4. Native ASCII contact and backing histories. The initial pusher peak includes acceleration of the initially stationary spreaders; it is not the steady foam load or a support-frame qualification. Values and sampling provenance are in the force-path data and contact histories.

5. Shell stress and deformation

Von Mises values below are initial-volume-weighted element percentiles maximized over saved states. P95 and P99 can peak at different times. The pad-root selection spans both shells at $112 \leq |X| \leq 150$ mm; the reported body excludes those elements. Stress is elastic demand, not a failure probability.

Region	Peak P95 (MPa)	Peak P99 (MPa)
Lower shell	72.72	107.31
Upper shell	72.98	107.75

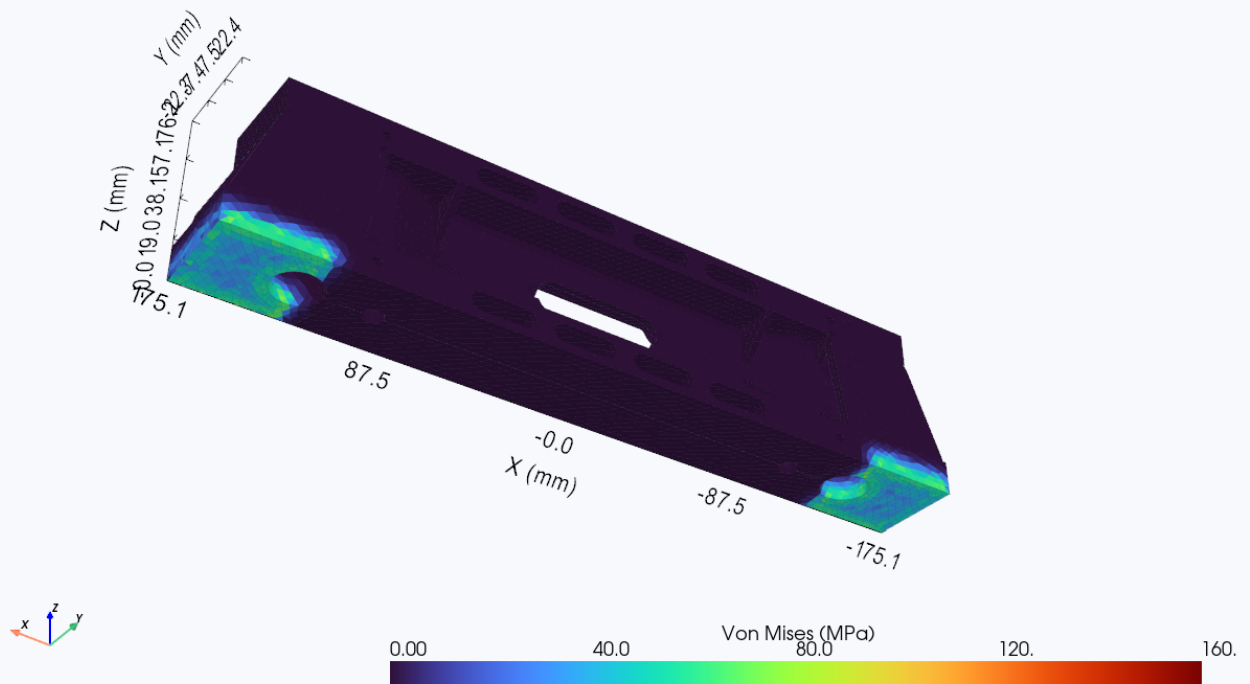
Region	Peak P95 (MPa)	Peak P99 (MPa)
Pad roots, both shells	93.93	119.82
Body excluding pad roots	54.18	100.77
Central span	59.19	75.75
All printed material	72.89	107.41



Stress and deformation history

Figure 5. Early pad/root response and later body loading are tracked separately. Nonrigid displacement removes a mass-weighted best-fit rigid motion from the whole two-shell pusher.

clamshell-4mm-v1 t = 0.010 ms
Rigid motion removed; deformation 1x; original outline gray



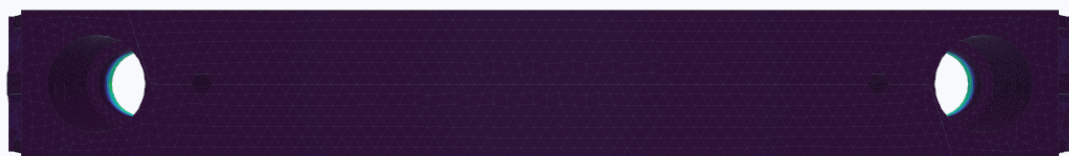
Display saturates at 160 MPa; raw peak 563.7 MPa elastic proxy

Early stress oblique

Figure 6. Early impact response at the saved frame nearest the native initial force peak.

clamshell-4mm-v1 t = 0.010 ms
Rigid motion removed; deformation 1x; original outline gray

TOP (+Z)



Display saturates at 160 MPa; raw peak 563.7 MPa elastic proxy

Early top view

clamshell-4mm-v1
Rigid motion removed; deformation 1x; original outline gray

t = 0.010 ms

IMPACT FACE (-Z)

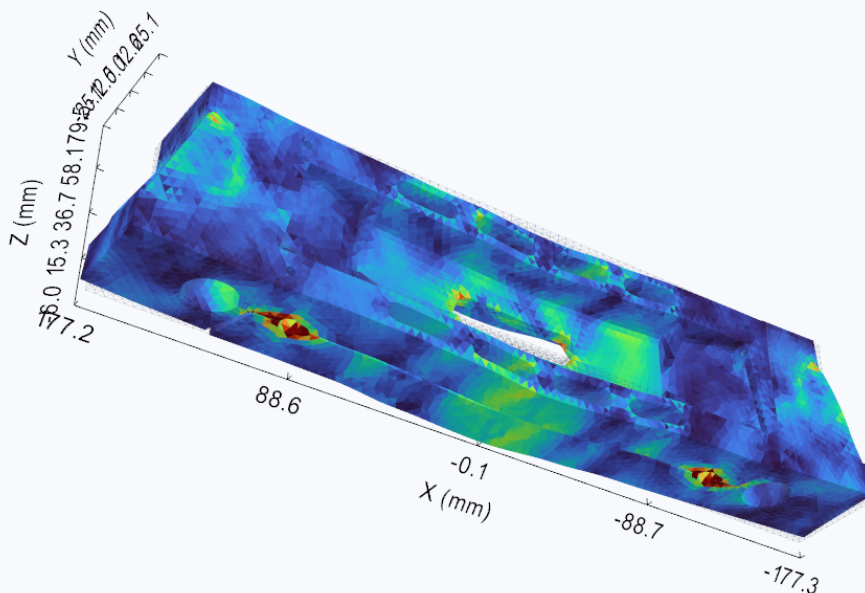


Display saturates at 160 MPa; raw peak 563.7 MPa elastic proxy

Early impact-face view

clamshell-4mm-v1
Rigid motion removed; deformation 1x; original outline gray

t = 0.390 ms



Display saturates at 160 MPa; raw peak 563.7 MPa elastic proxy

Later body stress oblique

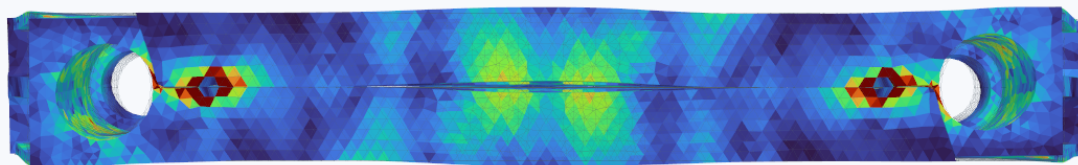
Figure 7. Broader shell/body loading near the body P95 peak. Both stress views use a fixed **0-160 MPa saturated turbo display**; values above 160 MPa share the highest color. The raw saved-state element maximum is 563.7 MPa, retained in the stress histories. This local maximum is not a mesh-independent

strength result.

clamshell-4mm-v1
Rigid motion removed; deformation 1x; original outline gray

t = 0.390 ms

TOP (+Z)



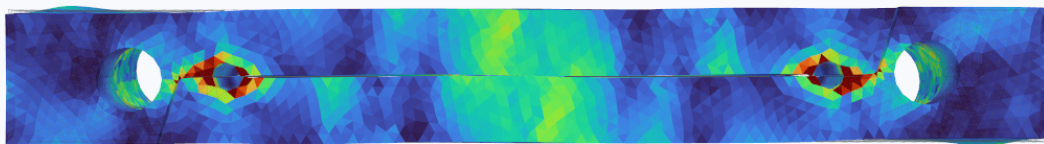
Display saturates at 160 MPa; raw peak 563.7 MPa elastic proxy

Later top view

clamshell-4mm-v1
Rigid motion removed; deformation 1x; original outline gray

t = 0.390 ms

IMPACT FACE (-Z)

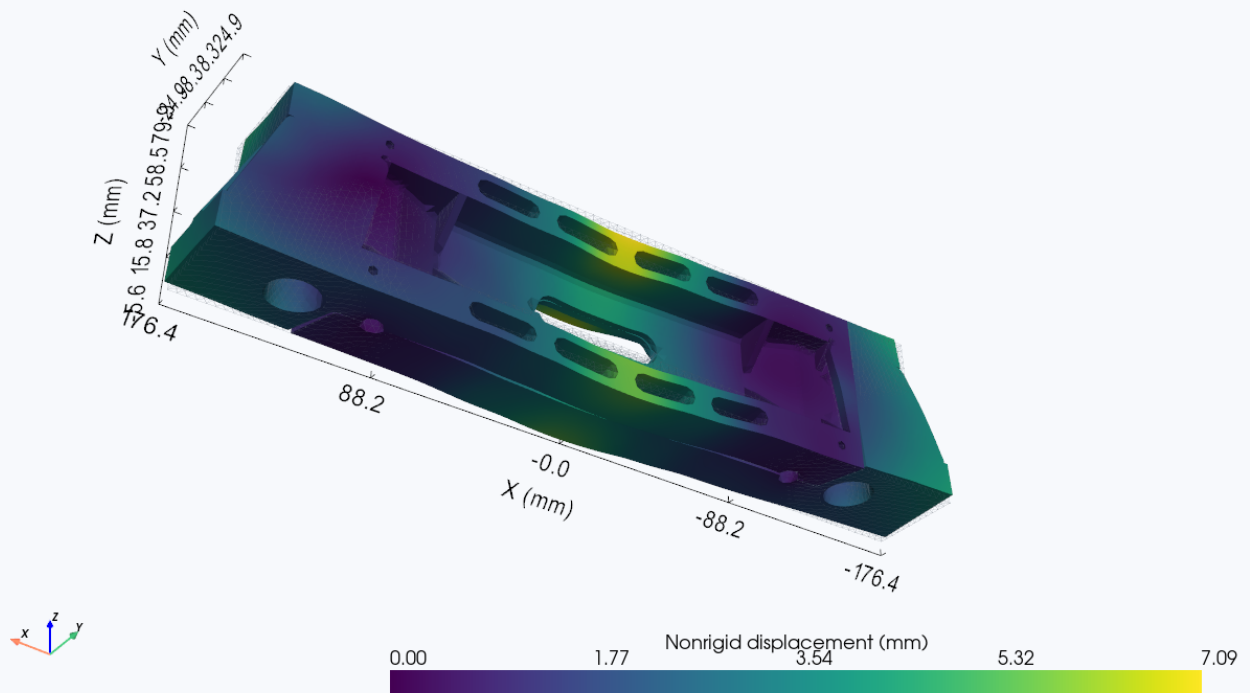


Display saturates at 160 MPa; raw peak 563.7 MPa elastic proxy

Later impact-face view

clamshell-4mm-v1
Rigid motion removed; deformation 1x; original outline gray

t = 0.420 ms



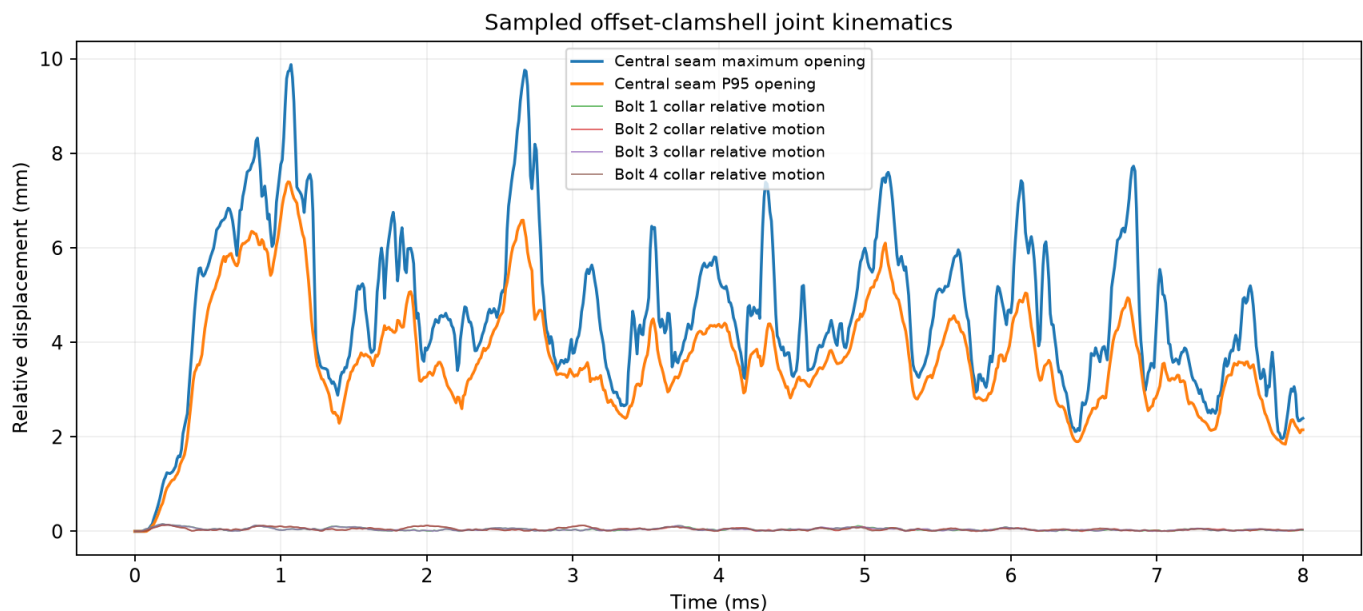
Elastic stress demand / generic foam — no failure prediction

Maximum nonrigid displacement

Figure 8. Largest nonrigid displacement at 0.420 ms. Maximum residual motion is 7.06 mm in Lower and 7.09 mm in Upper, measured relative to the combined pusher rigid fit. This differs from the overall stopping travel.

6. Offset seam and closure joints

The shells remain separate meshes with mortar contact across their mating surfaces. The four small nodal rigid collars idealize tightened closure bolts; they do not represent preload, fastener bending/stress, slip, bearing failure or opening at an actual threaded joint.



Sampled seam and bolt-collar motion

Figure 9. For nominal central $Y=0$ seam nodes within $|X| < 80$ mm, sampled Upper-minus-Lower relative Y motion reaches **9.88 mm opening** and 0.31 mm closure. The maximum difference between the mean motions of a paired bolt-collar group is 0.156 mm; that number is not physical bolt slip. The joint audit uses 820 nearest node pairs within 1.5 mm in initial XZ and saved binary states. It is a kinematic sample, not a native contact-gap or pressure bound.

7. Numerical evidence and qualification

The solver reached 8.000 ms with normal termination; the execution record and termination evidence are distinct from the single-case numerical assessment. Energy-balance error is 0.0133% of initial energy. Contact impulse and moving-momentum discrepancy is 0.0037%. Maximum summed absolute nonfriction contact energy is 1.385%. Added mass is 1.961 mg. The minimum saved-state foam Gauss-point relative Jacobian is 0.371; no sampled inversion or element deletion was detected.

The repaired 4 mm mesh contains 78,906 pusher tetrahedra. Its minimum tetrahedral altitude is 0.110 mm after 541 local contractions; no altitude is below 0.05 mm. Separate shells have no shared nodes, and both impact patches and four closure groups passed the pre-solve setup audit.

The saved-state planar pusher/plate penetration screen reaches 0.394 mm left and 0.403 mm right. The seam contact generated initial incident-contact detections; the largest reported overlap is 0.00118 mm. These sampled/initial values do not bound unsaved-cycle or curved-surface penetration.

Single-case check	Result
Normal termination	PASS
Requested time reached	PASS
Binary matches NODOUT	PASS
Moving mass consistent	PASS

Single-case check	Result
Initial energy error below 0.5%	PASS
Energy balance error below 5%	PASS
Hourglass below 5% initial energy	PASS
Sum abs nonfriction contact energy below 5% initial energy	PASS
Contact impulse momentum error below 5%	PASS
No axial SPC on moving or guide nodes	PASS
Rigid guides leave only axial translation free	PASS
No prescribed motion	PASS
No detected deletion	PASS
Positive sampled foam Jacobians	PASS
First stop captured	PASS
Foam compression below 80%	PASS
Mass scaling disabled or quantified below 0.1% moving mass	PASS
All transverse guide equations satisfied and no axial equation	PASS

The 4 → 3 mm candidate mesh screen **fails**. Failed measures: Bolt-collar mean relative motion. Detailed mesh comparison.

Measure	4 mm	3 mm	Change	Screen
Peak stopping force (kN)	251.546	255.259	1.48%	PASS ($\leq 5\%$)
First stop travel (mm)	122.646	122.770	0.10%	PASS ($\leq 5\%$)
Maximum nonrigid deformation (mm)	7.089	7.381	4.12%	PASS ($\leq 5\%$)
Lower shell P95 (MPa)	72.721	72.697	0.03%	PASS ($\leq 10\%$)
Upper shell P95 (MPa)	72.980	72.520	0.63%	PASS ($\leq 10\%$)
Central seam maximum opening (mm)	9.878	10.364	4.92%	PASS ($\leq 5\%$)

Measure	4 mm	3 mm	Change	Screen
Bolt-collar mean relative motion (mm)	0.156	0.248	58.37%	FAIL ($\leq 5\%$)

The refined case reached the full 8 ms with all its single-case numerical checks passing (3 mm metrics, 3 mm termination); its independent setup audit also passed. A full-event candidate timestep refinement has not been run; the reference's timestep result does not by itself qualify this new seam. The reference's targeted cable-cap thickness refinement fails its own 5% deformation screen, so an overall design ranking remains unqualified.

8. Same-fixture comparison with the solved 340 mm reference

The comparison reader verifies each run/deck hash, source CAD identity, actual material/foam/contact cards, guide definitions, plate geometry, equal 8 ms output windows and independent numerical checks. It accounts for the right impact contact moving from reference part 1 to this assembly's Upper part 2. The machine-readable comparison records any failed identity check and the raw responses. Its status is **same_fixture_response_comparison**; response differences below are observations, not a strength ranking.

Measure	340 mm reference	350 mm clamshell	Change
Printed pusher mass (g)	488.74	579.77	18.63%
Moving mass (g)	650.34	741.37	14.00%
Initial kinetic energy (J)	812.93	926.72	14.00%
Peak pusher-to-spreader force (kN)	174.10	251.55	44.48%
First stop time (ms)	5.427	5.273	-2.84%
First stop travel (mm)	113.98	122.65	7.61%
Maximum moving travel (mm)	113.98	122.65	7.61%
Left foam compression (%)	56.09	60.58	8.00%
Right foam compression (%)	56.12	60.55	7.90%
Global pusher P95 stress (MPa)	79.68	72.89	-8.53%
Global pusher P99 stress (MPa)	112.85	107.41	-4.82%

The printed mass includes the candidate's own two-shell volume and the same two motor masses. Differences in stopping, compression, contact force or global stress arise from geometry and model response under the frozen fixture. This does not establish which design survives or whether the four real closure bolts retain the shells.

9. Engineering interpretation and limits

The model supports a same-fixture comparison of stopping travel, load path and elastic demand **within its numerical checks**. The candidate has a physically meaningful offset seam in the model, and the sampled seam motion deserves joint-specific hardware review. A 50 m/s impact can generate a much larger short-duration pusher-to-spreader peak than the foam/backing peak because the spreaders begin at rest. Foam capacity and actual printed-part safety remain unverified.

The assumed PA12-CF10 isotropic elastic law omits print anisotropy, moisture, rate dependence and damage. The synthetic MAT057 foam curve is anchored at one catalog point, without high-rate product calibration. Ideal cable guides omit clearance, friction and cable compliance. Bolts lack geometry, preload and strength; seam contact stiffness and friction are assumed. Support-frame compliance, continuing motor thrust, fracture/deletion and repeated impact are absent. Mesh and timestep checks are limited as stated above. The fixed stress color range intentionally saturates above 160 MPa and must not be read as a failure threshold.

For reproduction and the full raw solver location, see run commands, source-code hashes, the candidate model and numerical metrics. The ZIP contains selected evidence; complete raw binary databases remain at the recorded case path.

Authored source: [projects/offset_clamshell_impact_20260922/report/REPORT.md](#). Original narrative and values retained. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.